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DETAIL SPECIFICATION

ETHYL CENTRALITE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for ethyl centralite for use in propellants and explosives.

1.2 Classification. The ethyl centralite is of the following Classes (see 6.1):

TABLE I. Classification.

Class 1	Lump
Class 2	Powder
Class 3	Flake

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM-AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil . Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1	-	Standard Specification for ASTM Liquid-in-Glass Thermometers
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM E969	-	Standard Specification for Glass Volumetric (Transfer) Pipets

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

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3.2 Physical and chemical properties. Ethyl centralite shall conform to the physical and chemical requirements specified in Table II.

TABLE II. Physical and chemical properties.

Property	Requirement
Solidification point, degrees Celsius (˚C)	71.0 to 72.5
Melted material appearance	Bright clear liquid, free from scum and deposit
Volatile content, weight percent maximum (max.)	0.1
Ash content, weight percent max.	0.1
Acidity, hydrochloric acid weight percent max.	0.04
Sum of secondary and tertiary amines (as ethylaniline and diethylaniline), weight percent max.	0.20
Hydrolyzable chlorine compounds, weight percent max.	0.001
Particle form Class 1: Class 2:Minimum weight percent passing through ASTM E11 Standard 600 micrometers (µm) Class 3:Minimum weight percent passing through ASTM E11 Standard 5.6 millimeters (mm)	Shall be supplied in lump form 99.9 99.9
Particle thickness Class 3:Average flake thickness, mm, max.	0.635

3.3 Workmanship. Ethyl centralite shall be free from grit, visible impurities, and foreign material such as dirt, rust, paint, wooden particles or any other foreign objects.

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4. VERIFICATION

TABLE III. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration		3 – Examination 4 – Test				Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Solidification point	X		X	X	X	X	4.4.1
3.2	Melted material appearance	X		X	X	X	X	4.4.2
3.2	Volatile content	X		X	X	X	X	4.4.3
3.2	Ash content	X		X	X	X	X	4.4.4
3.2	Acidity	X		X	X	X	X	4.4.5
3.2	Secondary and tertiary amines	X		X	X	X	X	4.4.6
3.2	Hydrolyzable chlorine compounds	X		X	X	X	X	4.4.7
3.2	Particle form	X		X	X	X	X	4.4.8
3.2	Particle thickness	X		X	X	X	X	4.4.9
3.3	Workmanship			X		X	X	4.5

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table IV. Sampling procedures shall be conducted in accordance with 4.3.2 or 4.3.3. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

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TABLE IV. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Solidification point	3.2	4.4.1
102	Melted material appearance	3.2	4.4.2
103	Volatile content	3.2	4.4.3
104	Ash content	3.2	4.4.4
105	Acidity	3.2	4.4.5
106	Secondary and tertiary amines	3.2	4.4.6
107	Hydrolyzable chlorine compounds	3.2	4.4.7
108	Particle form	3.2	4.4.8
109	Particle thickness	3.2	4.4.9
Minor			
201	Workmanship	3.3	4.5

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph “Formation and identification of lots or batches.” A lot shall consist of one or more acceptable batches of ethyl centralite produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions. Each batch shall consist of that quantity of ethyl centralite that has been subjected to the same chemical and physical unit operations intended to make the final product homogeneous.

4.3.2 Sampling at the manufacturer. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 550 grams (g) shall be formed which is assumed to be representative of that batch. All composite samples shall be verified in accordance with 4.3.4. Failure of a sample to comply with any of the requirements specified herein requires rejection of the batch represented by the sample.

4.3.3 Sampling at the receiving facility. Sampling shall be conducted in accordance with ASTM E300. Ethyl Centralite packaged in shipping containers (e.g., drums, boxes, etc.) shall be sampled using Verification Level II for attributes of MIL-STD-1916, where on unit is defined as one container. Each container shall be sampled in two locations using a thief stick. The samples shall then be placed in a secondary container and mixed to form a composite sample of not less than 550 g for each lot. There shall be no reduced sampling. Every lot received shall be tested in

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accordance with 4.3.4. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected.

4.3.4 Ethyl centralite.

Category	Examination or test	Verification level	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Solidification point	4.3	3.2	4.4.1
102	Melted material appearance	4.3	3.2	4.4.2
103	Volatile content	4.3	3.2	4.4.3
104	Ash content	4.3	3.2	4.4.4
105	Acidity	4.3	3.2	4.4.5
106	Secondary and tertiary amines	4.3	3.2	4.4.6
107	Hydrolyzable chlorine compounds	4.3	3.2	4.4.7
108	Particle form	4.3	3.2	4.4.8
109	Particle thickness	4.3	3.2	4.4.9
Minor				
201	Workmanship	4.3	3.3	4.5

4.4 Test methods and procedures. Unless otherwise specified, all chemicals used shall be at least reagent grade.

4.4.1 Solidification point. Fill the inner tube of the solidification point apparatus with a 10 to 12 cm high column of sample. Place the tube into a 90 to 100°C glycerin bath and allow the sample to melt. (A water bath can be used instead, however, stopper the test tube to prevent condensation of water onto the sample). Note the appearance of the molten material (See 4.4.2). Refill the tube with additional sample as necessary to get the liquid to the level such that the ASTM 93C-86 thermometer (76 mm partial immersion, 60-90°C range, 0.1°C increments in accordance with ASTM-E1) is at the proper immersion level, its tip is 3 to 5 cm off the bottom and the bulb is radially centered in the tube, and that the stirrer can sufficiently agitate the liquid. Stopper the test tube with the 2-hole stopper, fitted with the stirring rod with loop end and the thermometer. Remove the tube from the bath, wipe the outside of it clean and place the test tube into the solidification point apparatus (See Figure 1). Stir the molten material constantly, making approximately one up and down motion of the stirrer per second. The temperature of the molten material shall be recorded every 30 seconds until solidification begins as indicated by the temperature of the sample rising after it has reached a minimum. The stirring shall be continued until the temperature reaches a maximum. Report this maximum temperature as the solidification point. Immediately re-melt the material to facilitate its disposal.

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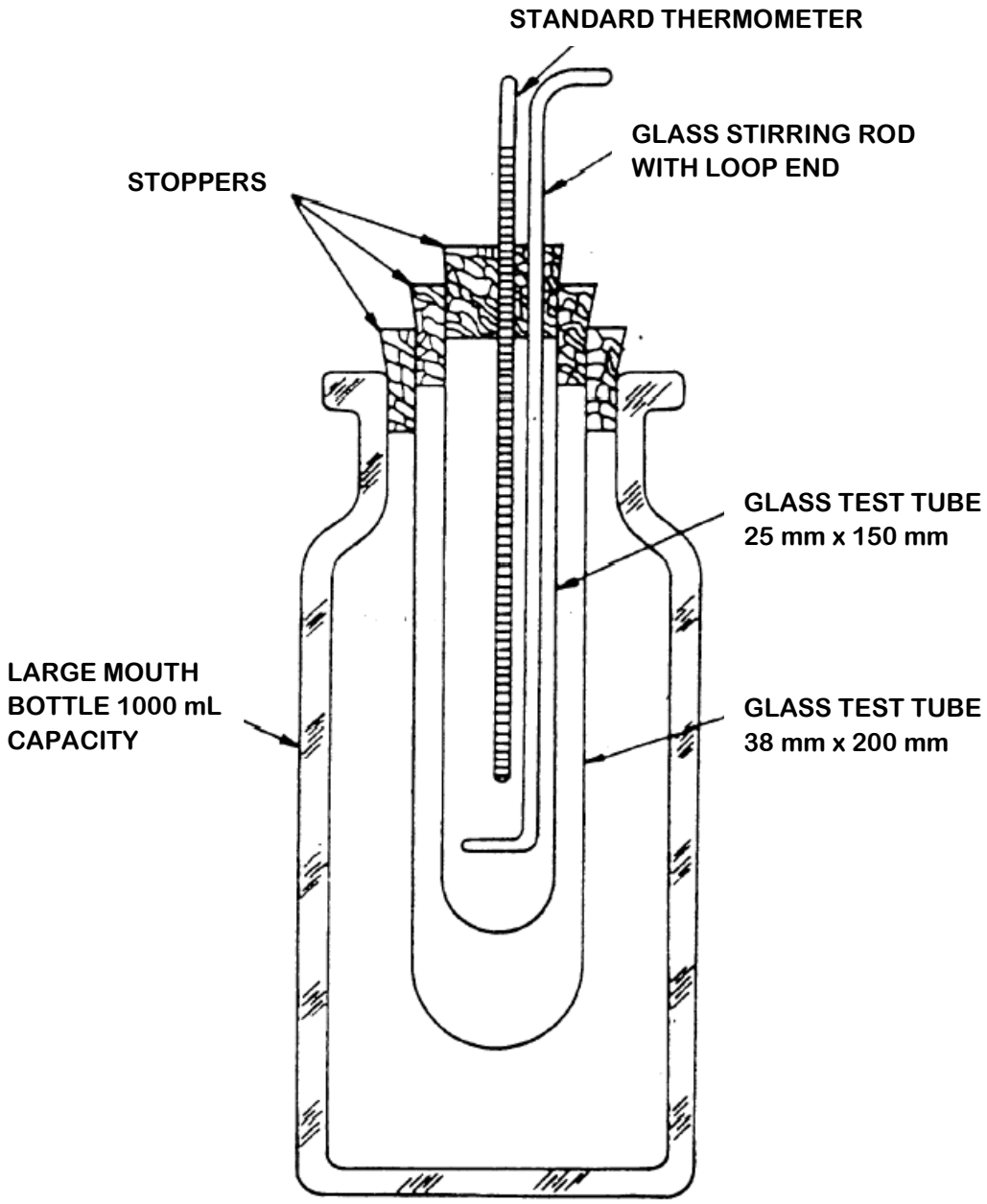


FIGURE 1. Solidification point apparatus.

4.4.2 Melted material. The appearance of the melted material shall be determined visually for compliance with the requirement during the determination of solidification point. The examination shall be made during the test specified in 4.4.1.

4.4.3 Volatile matter. A weighed portion of approximately 5.0 g, recorded to the nearest 0.1 g, of the sample shall be transferred to a shallow dish, the gross weight recorded to the nearest 0.0001g, the dish heated for two hours at 60˚C, cooled in a desiccator to room temperature and re-weighed. The volatile matter shall be calculated as follows:

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$$\text{Percent volatiles} = \frac{(A - B) \times 100}{W}$$

Where: A = gross weight of initial sample, g
B = gross weight of dried sample, g
W= sample weight, g

4.4.4 Ash. A portion of 5.0 ± 0.5 g of the sample shall be transferred to a previously ignited (at 600 ± 50° C) crucible pre-weighed to the nearest 0.0001 g. Sufficient alcohol (ethanol or isopropanol) shall be added to cover the material. The mixture shall be ignited, allowed to burn down slowly and then heated in a muffle furnace at 600 ± 50°C until all carbonaceous material has been removed. The crucible shall be cooled on a heat-proof surface until cool enough to be placed in a desiccator and further cooled to room temperature. Reweigh the crucible and ash to the nearest 0.0001g. The percent ash shall be calculated as follows:

$$\text{Percent ash} = \frac{(A - B) \times 100}{W}$$

Where: A = gross weight of crucible and ash, g
B = weight of empty crucible, g
W = sample weight, g

4.4.5 Acidity. A weighed portion of approximately 25 g, recorded to the nearest 0.1 g, of the sample shall be placed in a 250 mL Erlenmeyer flask and melted with 30 mL of distilled water at 90°C ± 5°C. The mixture shall be shaken vigorously until the ethyl centralite crystalizes. The mixture shall be cooled to room temperature and titrated with 0.1N sodium hydroxide solution, using phenolphthalein as an indicator. At the same time, a blank determination shall be run and the volume of sodium hydroxide used in titrating the sample corrected. Acidity shall be calculated as hydrochloric acid as follows:

$$\text{Percent hydrochloric acid} = \frac{3.65 \times (V - C) \times N}{W}$$

Where: V = mL of sodium hydroxide used in titration
C = mL of sodium hydroxide used in blank determination titration
N = normality of sodium hydroxide solution
W = weight of sample, g

4.4.6 Secondary and tertiary amines.

4.4.6.1 Secondary amines (ethylaniline). A weighed portion of approximately 25 g of the sample, recorded to the nearest 0.1 g, shall be transferred to an Erlenmeyer flask and dissolved with 30 to 50 mL of toluene. 5 mL of an approximately 0.25N solution of acetic anhydride in xylene shall be added. The flask shall then be stoppered and allowed to stand overnight. 25 mL of 0.1N sodium hydroxide solution shall be added, shaken thoroughly and transferred to a

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separatory funnel. The flask shall be washed with distilled water and the washings added to the contents of the funnel. 25 mL of 0.1N hydrochloric acid shall be added, shaken and the aqueous layer then separated. The funnel shall be washed with successive quantities of distilled water, adding the washings to the separated aqueous layer. The aqueous layer shall be titrated with 0.1N sodium hydroxide solution using the phenolphthalein as an indicator, until the first persistent pink color is observed. At the same time, a blank determination shall be run through the complete procedure. The percentage of ethylaniline shall be calculated as follows:

$$\text{Percent ethylaniline} = \frac{12.1 \text{ x N x } (B - (V - C))}{W}$$

Where: B = mL of sodium hydroxide solution required for blank
V = mL of sodium hydroxide solution required for sample
C = mL of sodium hydroxide required for acidity of the sample (see 4.4.5)
N = normality of sodium hydroxide solution
W = weight of sample, g

4.4.6.2 Tertiary amines (N,N-diethylaniline). A weighed portion of 5.00 ± 0.10 g of the sample shall be dissolved in 20 mL of toluene in a separatory funnel. A 20 mL aliquot of 0.5 M hydrochloric acid (e.g. 4.4 ± 0.2 mL of 34-37%wt hydrochloric acid diluted to 100 mL with deionized water) shall be added, and the mixture shaken thoroughly for at least 15 seconds. Rest the funnel in a ring stand and allow the layers to separate. Transfer the lower, aqueous layer to a 50 mL volumetric flask. Add 2 mL of 1.0 M sodium nitrite solution (e.g. 0.69 ± 0.01 g sodium nitrite diluted to 10 mL with deionized water) and sufficient deionized water to make the total volume up to 50 mL. The sample solution shall be thoroughly mixed and allowed to stand for 30 minutes. Prepare a 1.0 g/L stock solution of N,N-diethylaniline by dissolving 0.050 ± 0.005 g of it in 50 mL of toluene in a volumetric flask. A standard shall be prepared in the same manner as the sample except the 5 g sample shall be substituted with 1.00 mL of the 1.0 g/L stock solution of N,N-diethylaniline. A colorimeter or spectrophotometer equipped with a 10 mm pathlength cell shall be used to measure the absorbance of all solutions at 425 nm (nanometers). The instrument response shall be baselined (blanked) using deionized water. The absorbance of the sample solution shall be compared with that given by the standard. Calculate the percentage of tertiary amines (as diethylaniline) in the sample:

$$\text{Percent tertiary amines} = \frac{A_{smp} \text{ x } W_{std}}{10 \text{ x } A_{std} \text{ x } W_{smp}}$$

Where: A_{smp} = sample absorbance at 425 nm
A_{std} = standard absorbance at 425 nm
W_{smp} = weight of sample, g
W_{std} = weight of diethylaniline used to prepare stock solution, g

4.4.6.3 Sum of secondary and tertiary amines. The sum of the secondary and tertiary amines shall be reported for conformance.

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4.4.7 Hydrolyzable chlorine compounds.

4.4.7.1 Ion chromatography (preferred method). The following procedure is used to determine the amount of hydrolyzable chlorine compounds. The ethyl centralite is reacted with 5% sodium ethylate, the chloride ion extracted, and measured by ion chromatography with conductivity detection.

4.4.7.1.1 Reagents.

a. Deionized (DI) water, with conductivity below 0.1 microsiemens per centimeter (µS/cm), dispensable via squirt bottle.

b. 5% weight solution of sodium ethylate in ethanol, may be prepared by a 4-fold dilution in ethanol of a commercially-available 20% weight concentrate.

c. Phosphoric acid, 85% weight aqueous solution, American Chemical Society (ACS) grade.

d. Sodium chloride, ACS grade.

e. Chromatography eluent, an aqueous solution in DI water, typically containing 2 to 6 millimolar sodium carbonate and 1 to 2 millimolar sodium bicarbonate. The exact ratio depends on chromatography column (see 4.4.7.1.2.e); a low concentration of carbonate yields slower elution but better resolution, while a high concentration of carbonate yields faster elution but poorer resolution.

f. Suppressor regenerating solution as recommended by chromatograph manufacturer, typically a 50 to 100 millimolar solution of sulfuric acid in DI water.

g. Acetone, ACS grade.

4.4.7.1.2 Supplies and equipment.

a. Analytical balance, 200 g capacity, 0.001 g resolution or better.

b. Hot plate with adjustable surface temperature or steam bath.

c. Ultrasonic cleaning bath.

d. Ion chromatograph with anion suppression and conductivity detector with minimum sensitivity of 100 µS/cm full scale.

e. Ion chromatography column, which together with the proper eluent (4.4.7.1.1.e) can baseline-resolve chloride and completely elute phosphate in 15 minutes or less, e.g. Metrohm Metrosep ASupp 5 150 x 4 millimeters 5 micron, or equivalent.

f. Filtration supplies: funnel and filter paper (e.g. Whatman Number 41) for gravity filtration or 3 medium porosity sintered-glass bottom crucibles and vacuum pump and bell for vacuum-aided filtration.

g. Volumetric flasks, quantity: 2 x 1 liter (L), 3 x 100 mL, 1 x 10 mL.

h. Beaker, 150 mL, quantity: 3 or more.

i. Cover glasses, 6.5 centimeter diameter, quantity: 3.

j. Stirring rod, glass or plastic, with flat end.

k. Volumetric pipet, 10 mL, ASTM E969-02 Class A.

l. Micro-pipette, variable volume 50 – 250 microliter, and disposable plastic tips.

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m. Graduated cylinder, glass, “to deliver”, 50 mL capacity with 1 mL gradations.

4.4.7.1.3 Calibration standard. Prepare calibration standard as follows:

a. Weigh 1.650 ± 0.005 g of sodium chloride into a 1 L volumetric flask and make up to volume with DI water and mix well. This yields a 1000 milligram (mg) per liter chloride stock solution.

b. Transfer a 10 mL aliquot of the stock solution via volumetric pipette to a second 1 L volumetric flask. Make up to volume with DI and mix well. This yields a 10 mg per liter chloride calibration standard, equivalent to the 0.001% maximum allowed by the requirement in 3.2.

4.4.7.1.4 Sample preparation. The analysis steps a - j below shall be performed on duplicate 5 g aliquots of ethyl centralite sample and on a reagent blank. Prior to starting, all flasks, beakers, and stirring rod shall be rinsed at least 3 times with DI water, then acetone, and allowed to air dry. Filter funnels and filter paper or filter crucibles shall be rinsed with DI water prior to analysis.

a. Weigh 5.00 ± 0.05 g of ethyl centralite into a 150 mL beaker. Repeat with second replicate. To prepare the reagent blank, leave the beaker empty and go to the next step.

b. To each beaker, add 10 mL of 5% weight sodium ethylate, cover with a watch glass, and heat the solution for 20 minutes on a steam bath or atop a hotplate with a surface temperature of 120 to 130°C.

c. Remove beakers from heat and quickly add 50 mL of DI water.

d. Using a micro-pipette, add 250 microliters of phosphoric acid to each beaker and mix with stirring rod.

e. Allow the solution to cool to room temperature and for the sample to crystallize for at least one hour.

f. Break up any large crystals and agglomerates with flat end of stirring rod and with the aid of sonication.

g. Decant and filter the supernatant solutions into 3 100 mL volumetric flask using the pre-rinsed filtration set-up (4.4.7.1.2.f).

h. To the residue in each beaker add 10 mL of DI water, and again mix, mash, sonicate, and filter (steps f and g).

i. Repeat previous step (step h) once more then wash down each filter with a few milliliters of DI water and catch the filtrates in the corresponding flasks.

j. Fill the 100 mL volumetric flask to volume with DI water and mix well. The steps a-j shall yield two duplicate samples and a reagent blank.

k. Prepare a spiked chloride sample. To a 10 mL volumetric flask, add 100 microliters of chloride stock solution prepared in 4.4.7.1.3.a. Fill up the flask to volume with one of the non-blank sample solutions prepared in previous step (step j). This yields a spiked sample whose chloride concentration has been increased by 10 mg per liter and can be used to ascertain the location of the chloride peak in the chromatogram (4.4.7.1.5).

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4.4.7.1.5 Procedure. Set up the ion chromatograph in accordance with manufacturer instructions. The instrument shall be equilibrated until a steady, linear baseline is obtained.

a. Fill up the appropriate instrument vials, one each, with the 10 mg per liter standard (see 4.4.7.1.3.b), the reagent blank, 2 sample solutions, and spiked sample solution (4.4.7.1.4.j-k).

b. All solutions shall be analyzed using method appropriate for the selected chromatography column (see 4.4.7.1.1.e and 4.4.7.1.2.d-e) and system. Allow each run to be just long enough to completely elute the phosphate anion.

c. Identify the chloride peak, which elutes during the first half of each run. For the standard, chloride shall be the largest peak in the chromatogram; and note its retention time. For the samples, examine the same retention time ± 5%, and compare the original and spiked samples prepared in 4.4.7.1.4.j-k; chloride shall be identified as the single peak that increases in height between original and corresponding spiked solution. For the reagent blank and the two samples, chloride shall be identified as the largest peak at the same retention time as the spiked sample within a ±2% retention time window.

d. Integrate each chromatogram using valley-to-valley type integration making sure the start and end points of the chloride peak both coincide with the local baseline minima. Obtain the chloride peak areas for each of the runs.

e. Calculate the reagent-blank-corrected, weight-normalized chloride peak areas for each sample as follows:

$$A_1 = \frac{5 \text{ x } \left(A_{1,RAW} - A_{RB}\right)}{W_1}$$

$$A_2 = \frac{5 \text{ x } \left(A_{2,RAW} - A_{RB}\right)}{W_2}$$

Where:
A_1 = corrected chloride peak area for first sample replicate
A_2 = corrected chloride peak area for second sample replicate
$A_{1,RAW}$ = raw chloride peak area for first sample replicate
$A_{2,RAW}$ = raw chloride peak area for second sample replicate
A_{RB} = raw chloride peak area for reagent blank
W_1 = weight of first sample replicate, grams
W_2 = weight of second sample replicate, grams

f. Find the arithmetic average of A_1 and A_2. A_1 and A_2 shall not differ from their average by more than 8%. A larger value indicates poor chloride extraction, in which case new duplicate samples shall be prepared per 4.4.7.1.4, paying particular attention to steps f – i thereof.

g. Compare the average area obtained in previous step f with the chloride peak area of the 10 mg per liter standard found in step d. For the sample to pass specification, the average area shall be less than the standard area.

4.4.7.2 Alternate method. A 5 g portion of the sample shall be transferred to a 150 mL beaker. 10 mL of a 5% solution of sodium ethylate in alcohol shall be added and the solution

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heated on a steam bath for fifteen minutes. 75 mL of distilled water shall be added to the contents of the beaker and sufficient time allowed for the ethyl centralite to crystalize. The solution shall be made acid to litmus with concentrated nitric acid and then filtered through a Number 41 Whatman filter paper, or equivalent, into a 100 mL Nessler tube. If not clear, the solution shall be refiltered. The beaker and filter paper shall be washed with distilled water until the Nessler tube attains a volume of 100 mL. 5 mL of 10% silver nitrate shall be added, mixed thoroughly, and the solution shall be allowed to stand for one-half hour. Concurrent with the analysis, standard solutions (containing 0.001% and 0.002% sodium chloride) shall be prepared as follows: 1.650 ± 0.005 g of reagent grade sodium chloride shall be transferred to a 1 L volumetric flask. The flask shall be filled to the mark with distilled water and mixed well. A 10 mL aliquot shall be transferred to a second 1 L flask, and also, a 20 mL aliquot shall be transferred to another 1 L volumetric flask, brought to the mark with distilled water and mixed well. 100 mL of each chloride standard shall be transferred to separate Nessler tubes. A third tube shall contain 10 mL of distilled water as a blank. Each standard solution in the Nessler tubes shall be acidified (to litmus paper) with concentrated nitric acid, and 5 mL of 10% silver nitrate solution shall be added to each tube, mixed well, and allowed to stand for 30 minutes. The turbidity of the sample mixture shall be compared with the standards by viewing downward against a well-lighted white background. The turbidity of the sample shall be less than that of the 0.001% standard solution.

4.4.8 Particle form.

4.4.8.1 Class 1, lump. Visually examine the sample for conformance.

4.4.8.2 Class 2, powder. A 6 inch diameter ASTM E-11 Standard 600 µm sieve shall be weighed to the nearest 0.01 g, and a bottom pan shall be attached. A weighed portion of 100 ± 2 g of sample shall be placed on the sieve, covered, and shaken for 10 minutes by hand or five minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. The sieve shall be carefully separated from the cover and the bottom pan, and weighed. The amount passing through the sieve shall be calculated as follows:

$$\text{Percent Thru} = 100 \times \frac{W + S - R}{W}$$

Where:

W = weight of sample, g
S = weight of empty sieve, g
R = weight of sieve and sample retained on sieve, g

4.4.8.3 Class 3, flake. The procedure in 4.4.8.2 shall be followed except the sieve shall be a 6 inch diameter ASTM E-11 Standard 5.6 mm.

4.4.9 Class 3, particle thickness. The average maximum thickness of 25 flakes shall be determined by use of a suitable gage.

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4.5 Workmanship. Spread approximately 300 g of ethyl centralite material out on a clean, white surface. The sample shall be visibly inspected and be uniform in quality and free from lumps (class 2 and 3), grit, impurities or foreign matter. Failure to meet the workmanship requirement during inspection or production shall result in batch rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Ethyl centralite is intended for use as a stabilizer and burn rate modifier in propellants.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
c. Provisions for submission/resubmission of first article samples.
d. Acceptance and description sheet - Acceptance and description sheets should be prepared for each lot in accordance with MIL-STD-1171.
e. Quantity required and delivery schedules.
f. Class required (see 1.2).
g. Levels of preservation and packing (see 5.1).

6.3 Consideration of data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, contract data requirements list, acceptance and description sheets (for explosives) are required for each lot. These sheets should be prepared and submitted by the contractor in accordance with MIL-STD-1171 or otherwise as directed by the contract.

6.4 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: CCDC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

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6.5 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to the technical agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency responsible for preparation of this specification. (See 6.5)

6.7 Incorporated changes. ECP Y21Q2001 provides documentation for noted changes.

6.8 Requalification of energetic formulation. The procuring activity should note when a first article is required for ethyl centralite. The energetic formulation it is used in may require requalification to demonstrate that it meets the requirements of AOP-7 and STANAG 4170. Contact the technical agency (see 6.4) for guidance on additional sample size and applicable qualification authority.

6.9 Subject term (key word) listing.

Propellant
Stabilizer
Burn rate modifier

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING INFORMATION

Custodians:
Army-AR
Navy-OS
Air Force-70

Preparing activity:
Army-AR

(Project number 6810-2020-013)

Review Activities:
Army-AV, CR, EA, MI, MR, TE
Navy-AS, SH
Air Force-11

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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